

Trade Agency Sandbox Portal — Feature Report

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1. Overview

The **Trade Agency Sandbox Portal** is a full-stack advisory intelligence platform built for Canadian export trade agencies. It enables trade advisors to manage a portfolio of Small and Medium-sized Enterprises (SMEs), assess their export readiness across nine standardized pillars, and provide actionable intelligence on target markets — all within a single, premium dark-themed workspace.

Architecture at a Glance

Next.js App Router (React 19 · Turbopack · TypeScript)	
Client Components	API Routes (Route Handlers)
AgencyPortfolio	/api/sme
ReportWorkspace	/api/assessment
ScoreGauge	/api/sandbox/comtrade
TradeIntelDash	/api/sandbox/tariffs
LandedCostSolvr	/api/sandbox/freight
MarginValidator	/api/sandbox/rates
SanctionsScreen	/api/sandbox/screen
RoadmapTimeline	/api/roadmap
AdvisorNotes	/api/advisor-notes
EdcCountryRisk	/api/report/pdf
CountryPlaybook	/api/health
Data Layer: Supabase / CSV-DB / In-Memory Store	
External: UN Comtrade · SEMA Sanctions List	

The portal follows a **two-page** navigation model:

Route	Purpose
/sandbox/agency	Portfolio dashboard — lists all SMEs, shows aggregate readiness
/sandbox/agency/report?id=<smeId>	Deep-dive report workspace with tabbed sections

2. Feature-by-Feature Breakdown

2.1 Portfolio Dashboard

Route: `/sandbox/agency`

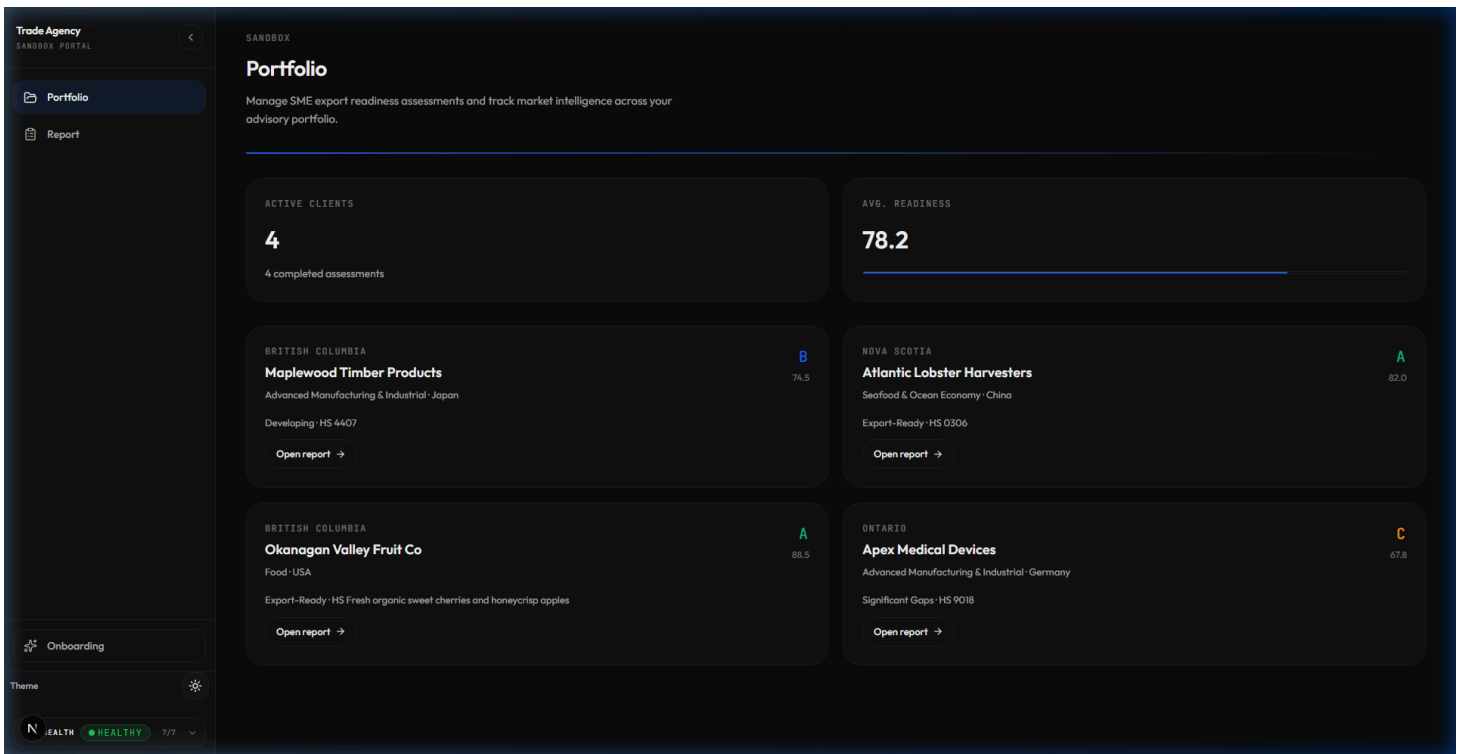
Business Problem: Trade advisors manage multiple SME clients simultaneously. They need a single view to triage which companies are export-ready, which have pending assessments, and the overall health of their advisory portfolio.

What it does:

- Displays aggregate KPIs: **Active Clients** count and **Avg. Readiness** score with animated progress bar
- Lists each SME in a bento-grid card layout showing:
 - Company name, province, industry, and target country
 - Readiness grade (A/B/C/D/F) with color-coded indicator
 - "Open report" link for assessed SMEs; inline "Start questionnaire" for pending ones
- Shows a real-time health status badge in the sidebar footer (7/7 services healthy)
- Sidebar navigation with Portfolio, Report, and Onboarding links plus theme toggle

Technical Details:

- Component: `AgencyPortfolioClient.tsx` — client-side RSC with `useState` / `useEffect` for data fetching
- Data: Fetches from `/api/sme` and `/api/assessment` in parallel via `Promise.all`
- Animation: `motion` (Framer Motion v12) staggered card entrance with `snappy` preset
- Scoring: `gradeColor()` and `gradeLabel()` from `lib/scoring-engine.ts` for grade visualization
- Inline questionnaire wizard: `QuestionnaireWizard` component can be launched directly from a portfolio card



2.2 Readiness Score Gauge

Business Problem: Advisors need an immediate, at-a-glance indicator of an SME's overall export readiness. A single number or letter grade isn't enough — the visual encoding must convey severity.

What it does:

- Animated SVG circular gauge with the readiness score (0–100) and letter grade (A–F)
- Color-coded ring: green for A, blue for B, amber for C, red for D/F
- Count-up animation on mount for the numeric score
- Grade label below (e.g., "Developing", "Export-Ready", "Significant Gaps")

Technical Details:

- Component: `ScoreGauge.tsx` — pure SVG with `motion.circle` for animated `strokeDashoffset`
- Custom hook: `useCountUp()` for animated decimal counter
- Constants: `SIZE=168` , `STROKE=10` , calculated `CIRCUMFERENCE` for arc math
- Transition: Uses `snappy` animation preset (critically-damped spring)

2.3 Sanctions & Compliance Screen

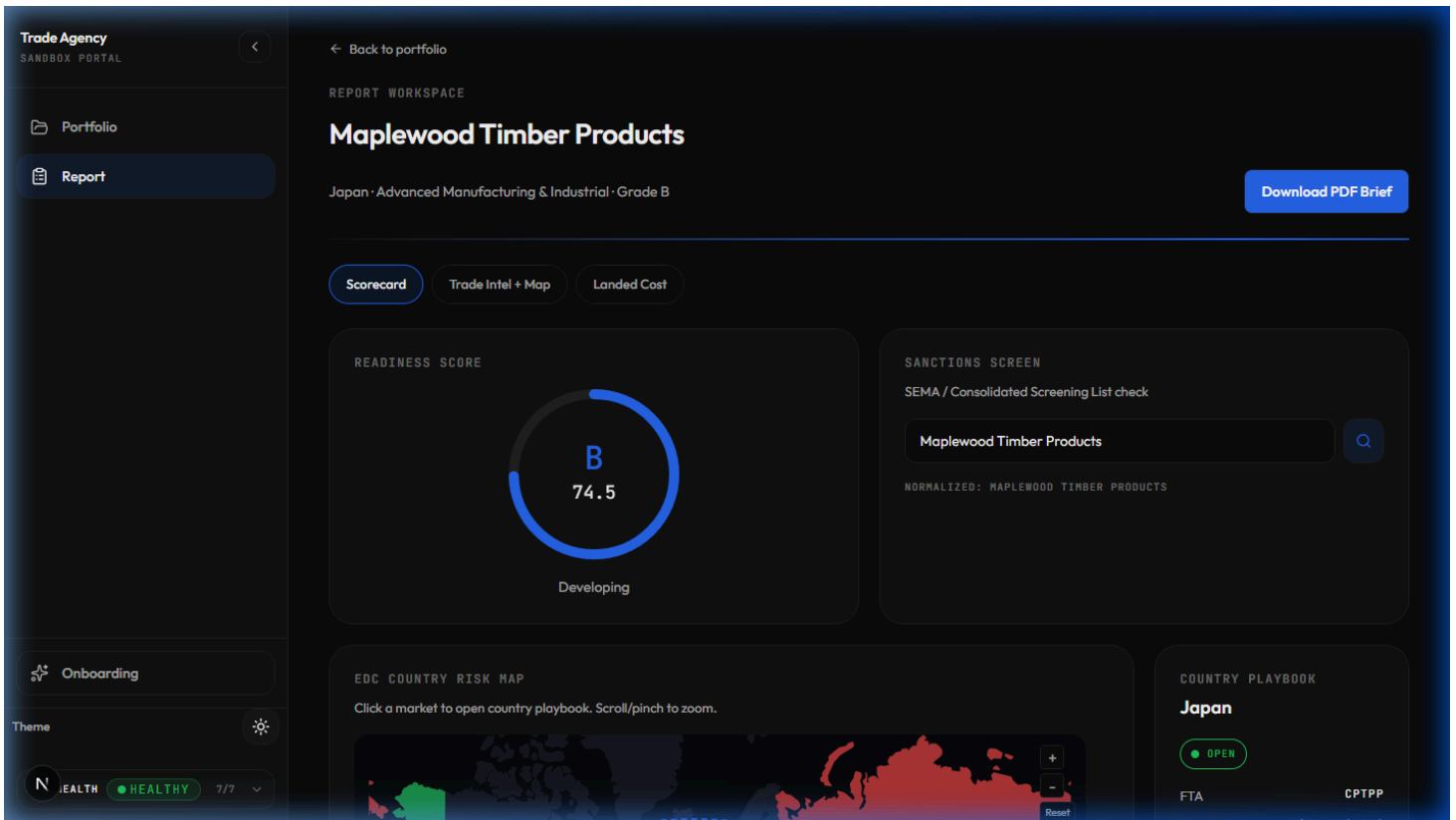
Business Problem: Before proceeding with any export engagement, advisors must verify the SME isn't on any sanctions or restricted entity lists. Failing to screen could result in severe legal penalties.

What it does:

- Text input pre-filled with the SME's name, with search button
- Normalizes the query (uppercase, trimmed) for consistent matching
- Calls `/api/sandbox/screen` to check against SEMA / Consolidated Screening List
- Shows "Clear" badge (green shield) or "Match Found" alert (red triangle) with matched entries
- Displays data origin (live/cache/mock-fallback) for audit trail

Technical Details:

- Component: `SanctionsScreen.tsx`
- Backend: `lib/sanctions-checker.ts` — `normalizeQuery()` for consistent name matching
- API: `/api/sandbox/screen` checks against structured sanctions data
- Animation: `motion.button` with `whileTap` scale and `motion.div` for result reveal



2.4 EDC Country Risk Map & Country Playbook

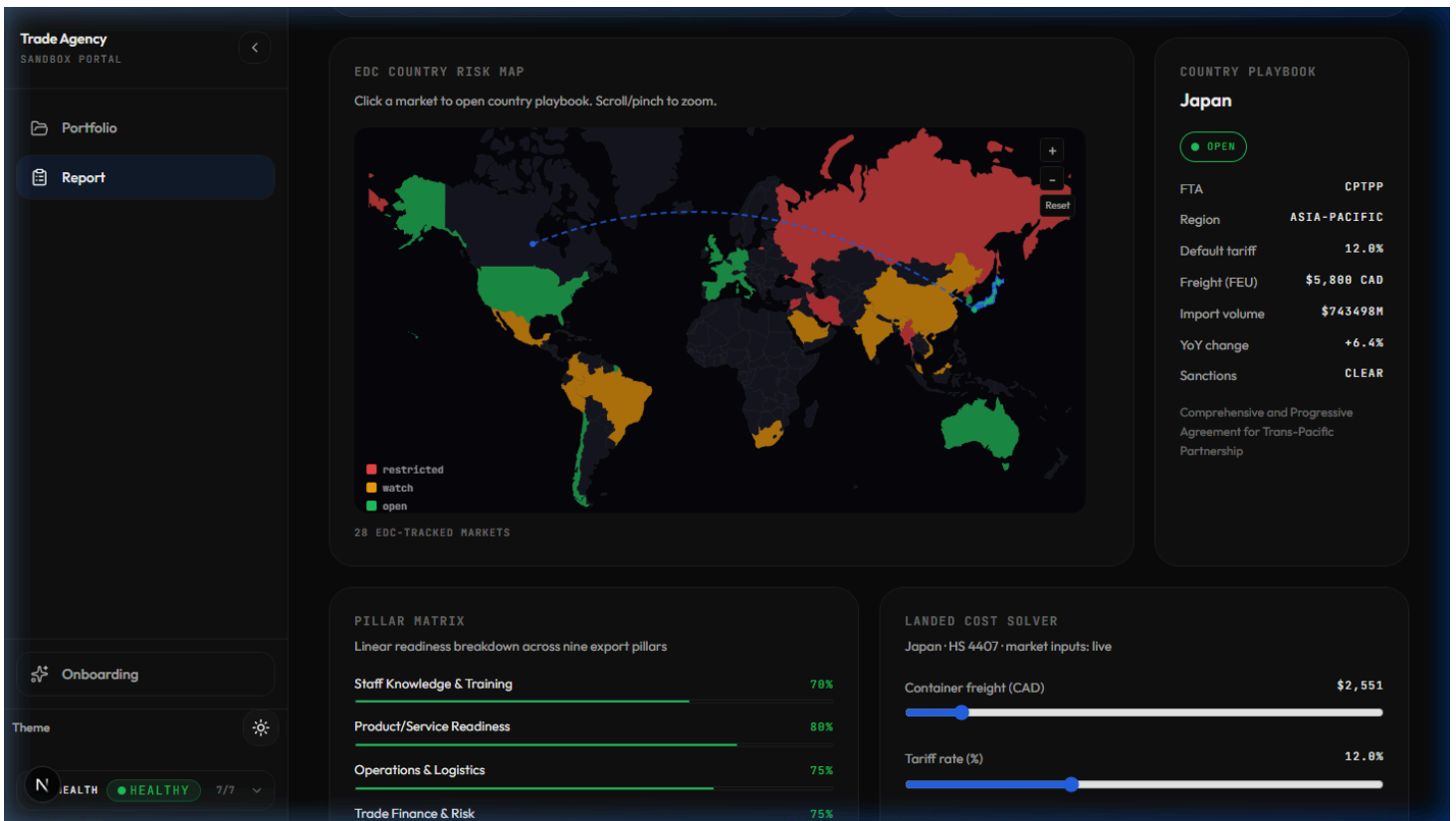
Business Problem: Export risk isn't just about the product — it's about the destination. Advisors need to visualize which markets are open, restricted, or blocked according to Export Development Canada's risk classifications, and quickly access market intelligence for any country.

What it does:

- **Risk Map**: Interactive Mercator world map colored by EDC risk tier:
 -  **Open** — low-risk, full coverage available
 -  **Watch** — elevated risk, selective coverage
 -  **Restricted / Blocked** — high risk, limited or no EDC coverage
- Animated dashed shipping lane arc from Canada to the selected target country
- Zoom/pan controls (scroll + buttons), reset functionality
- **Country Playbook** side panel: When a country is clicked, shows:
 - FTA status (e.g., CPTPP for Japan), region, default tariff rate
 - Freight cost per FEU in CAD, import volume, YoY change
 - Sanctions clearance status, FTA notes

Technical Details:

- Map: `@visx/geo` Mercator projection + `@visx/zoom` for pan/zoom
- Topology: `world-atlas@2` (110m resolution) loaded from CDN, parsed with `topojson-client`
- Risk data: `lib/country-risk-data.ts` — 28 EDC-tracked markets with ISO3 codes and numeric IDs
- Fallback: `buildMinimalWorld()` generates simple polygons if CDN fetch fails
- Shipping lane: SVG quadratic Bézier curve (`Q` path command) between Canada centroid and target country centroid
- Playbook: `CountryPlaybook.tsx` + `lib/mock-fallback-data.ts` for market profiles



2.5 Pillars Matrix

Business Problem: A single aggregate readiness score hides the nuance. Advisors need to see exactly which of the nine export readiness pillars are strong vs weak to guide their advisory recommendations.

What it does:

- Displays nine export pillars as horizontal animated bar charts:
 - i. Staff Knowledge & Training
 - ii. Product/Service Readiness
 - iii. Operations & Logistics
 - iv. Trade Finance & Risk
 - v. Legal & Regulatory
 - vi. Strategy & Market Selection
 - vii. Cultural Readiness
 - viii. Digital & E-Commerce
 - ix. Program & Funding
- Color-coded by score: $\geq 70\%$ green, $\geq 50\%$ blue, $\geq 25\%$ amber, $< 25\%$ red
- Animated bar width with `motion.div` spring transition

Technical Details:

- Component: PillarsMatrix.tsx
- Data: Record<PillarKey, number> from the assessment record
- Labels: PILLAR_LABELS from lib/scoring-engine.ts
- Layout: CSS flex column with progress bars using progress-track / progress-fill classes

2.6 Critical Gap Analyzer

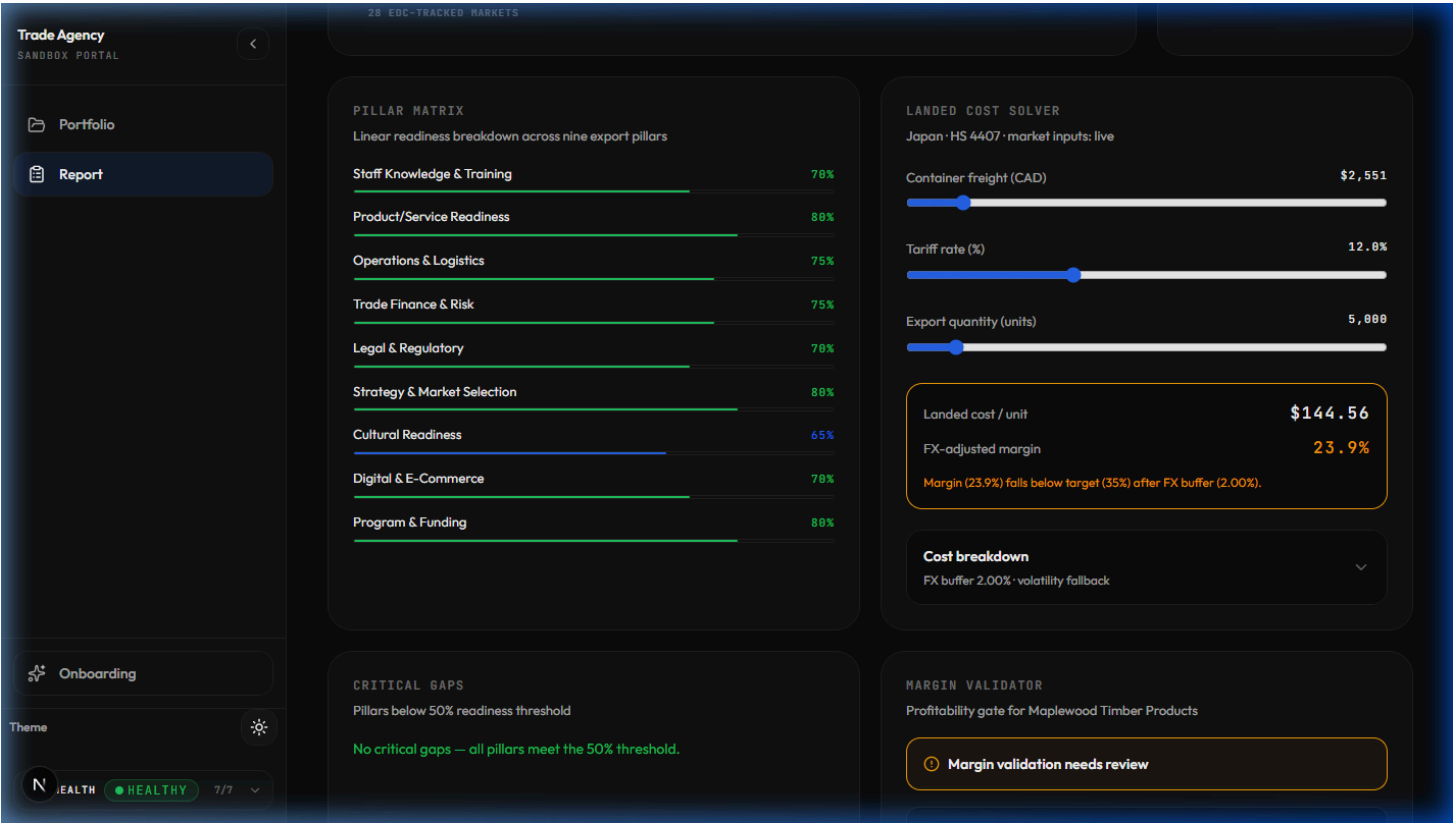
Business Problem: Advisors need to quickly identify which pillars fall below the readiness threshold (default 50%) so they can prioritize remediation efforts.

What it does:

- Filters pillars below the configurable threshold (default 50%)
- Lists them sorted by score (lowest first) with red percentage badges
- If no gaps exist, shows a green "No critical gaps" message

Technical Details:

- Component: CriticalGapAnalyzer.tsx
- Props: pillarScores + optional threshold (default 50)
- Staggered entrance animation with 40ms delay per gap item



2.7 Landed Cost Solver

Business Problem: Understanding whether an export deal is profitable requires calculating the total "landed cost" — production + freight + tariffs + insurance + brokerage + FX buffer. Advisors need an interactive tool to model different scenarios.

What it does:

- Three interactive sliders for real-time scenario modeling:
 - **Container freight (CAD):** 1,500–12,000 range
 - **Tariff rate (%):** 0% – 35% range
 - **Export quantity (units):** 500 – 50,000 range
- Results panel showing:
 - **Landed cost per unit** (e.g., \$144.56)
 - **FX-adjusted margin** (e.g., 23.9%) with color-coded border (green = pass, amber = near, red = insolvent)
 - Warning text when margin falls below target
- Expandable cost breakdown disclosure panel:
 - Unit freight, tariff per unit, broker fee, insurance, actual margin, target margin

Technical Details:

- Component: `LandedCostSolver.tsx`
- Calculator: `lib/landed-cost-calculator.ts` — `calculateLandedCost()` with FX volatility buffer
- Market inputs: Fetches live data from three APIs in parallel:
 - `/api/sandbox/freight` — container freight rates by origin/destination
 - `/api/sandbox/tariffs` — HS code tariff lookup by country
 - `/api/sandbox/rates` — FX volatility (30d/90d) for currency hedging
- Fallback: `lib/mock-fallback-data.ts` — `getCountryFallback()` for structured fallback when APIs fail
- FX Buffer: Uses 30-day or 90-day volatility to compute a currency risk buffer
- Data origin indicator: Shows `live`, `mock-fallback`, or `structured-fallback`
- UI: `SliderField` sub-component for range inputs, `FolderDisclosure` for expandable breakdown

2.8 Margin Validator

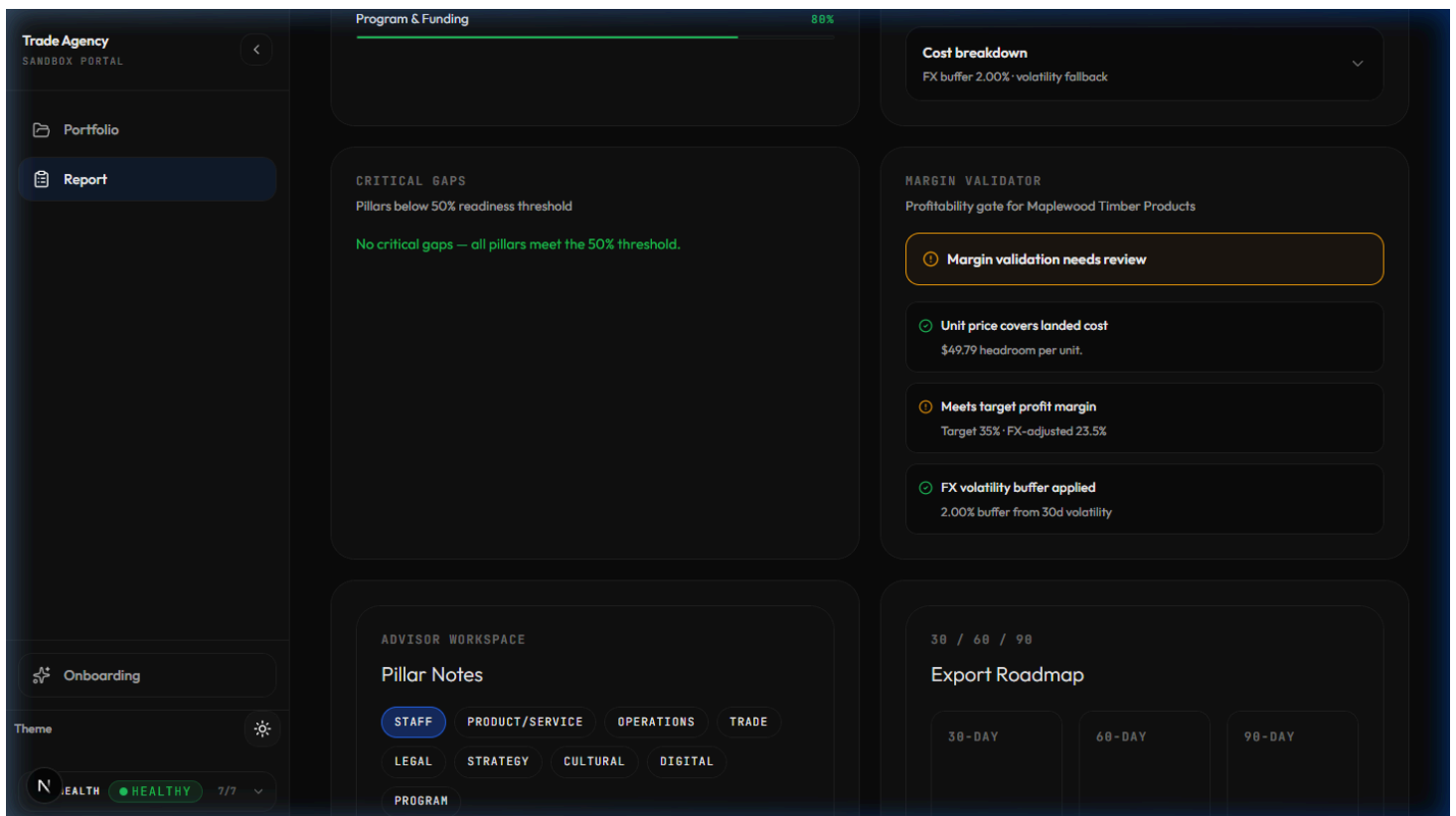
Business Problem: Even if the landed cost is calculated, advisors need a pass/fail gate to confirm the deal is profitable. Three key checks must all pass for an export deal to be viable.

What it does:

- Three validation checks, each with pass  / fail  indicator:
 - i. **Unit price covers landed cost** — headroom per unit
 - ii. **Meets target profit margin** — target vs FX-adjusted margin comparison
 - iii. **FX volatility buffer applied** — confirms buffer percentage from 30d/90d volatility
- Summary badge: "Margin validation passed" (green) or "Margin validation needs review" (amber)

Technical Details:

- Component: `MarginValidator.tsx`
- Reuses: `calculateLandedCost()` from `lib/landed-cost-calculator.ts`
- Props: Accepts optional pre-computed `LandedCostResult` to avoid duplicate calculation
- Icons: `CheckCircle2` and `AlertCircle` from `lucide-react`



2.9 Advisor Notes Panel

Business Problem: Advisors accumulate knowledge about each SME over time — notes per pillar, observations, recommendations. This needs to be persisted, organized by pillar, and auto-saved to avoid data loss.

What it does:

- Nine pillar tabs (Staff, Product/Service, Operations, Trade, Legal, Strategy, Cultural, Digital, Program)
- Textarea for free-form notes per pillar
- **Auto-save:** Notes are debounced (700ms delay) and persisted via `PUT /api/advisor-notes`
- Timestamp display: "Saved 01:41:58 · Assessment ASM-0000"

Technical Details:

- Component: `AdvisorNotesPanel.tsx`
- State: `Record<string, string>` map of pillar → content
- Auto-save: `useEffect` with `window.setTimeout` `debounce` (700ms)
- API: `GET /api/advisor-notes?assessmentId=...` (load), `PUT /api/advisor-notes` (persist)
- UX: `motion.button` with `whileTap` for pillar tab interactions

2.10 Roadmap Timeline (30/60/90)

Business Problem: Export readiness isn't instant — it requires a phased action plan. Advisors need to organize remediation tasks into 30-day, 60-day, and 90-day buckets, and reorder priorities via drag-and-drop.

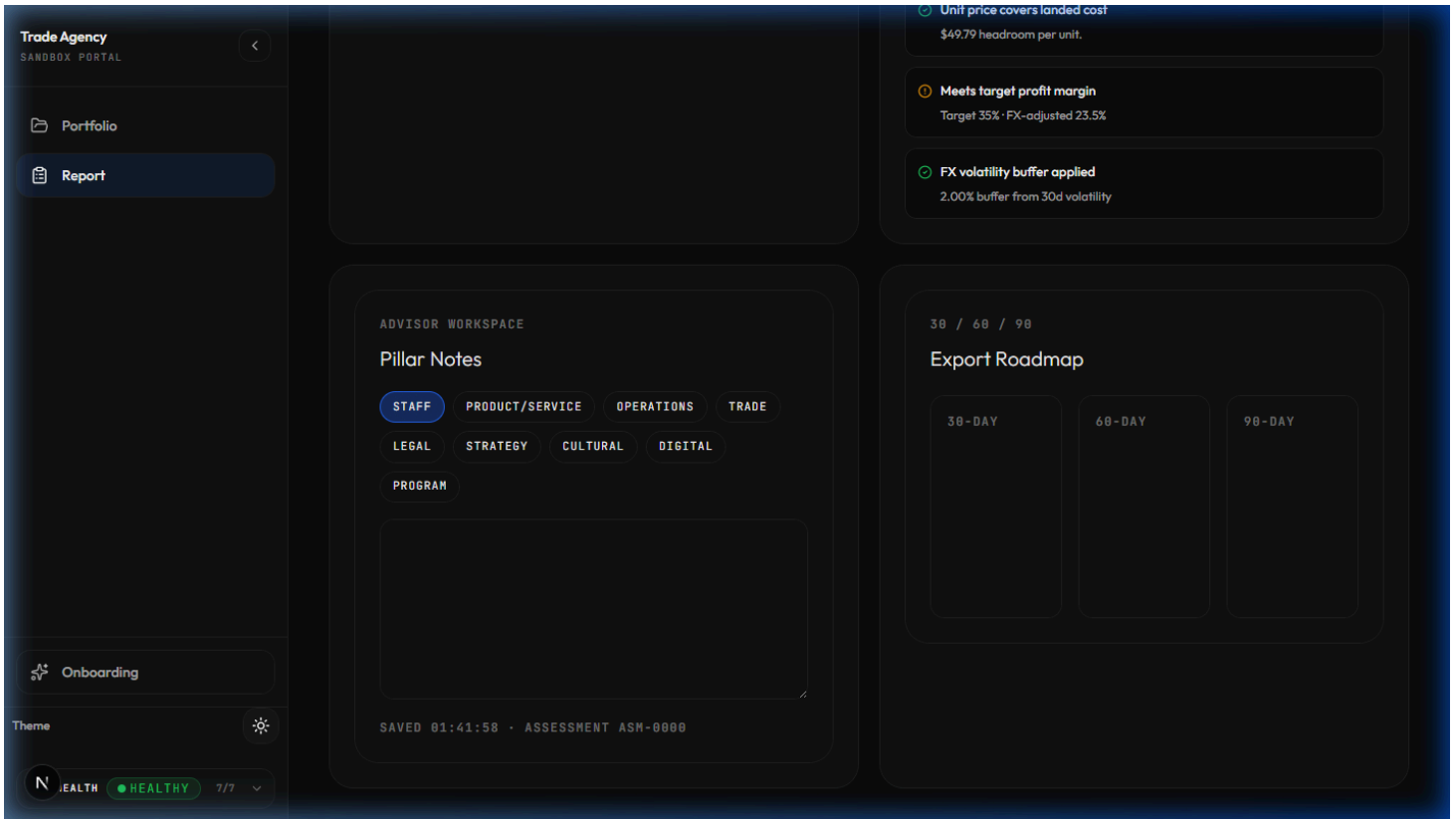
What it does:

- Three-column Kanban-style board: 30-day, 60-day, 90-day buckets
- Drag-and-drop cards between columns and within columns
- Each card shows a remediation task
- Reordering is persisted via `PATCH /api/roadmap`

Technical Details:

- Component: `RoadmapTimeline.tsx`
- Drag-and-drop: `@dnd-kit/core` + `@dnd-kit/sortable` with `PointerSensor` (6px activation constraint)
- Sub-component: `SortableCard` using `useSortable` hook with CSS transform

- DragOverlay : Premium styled overlay card during drag
- Persistence: arrayMove() for reordering, then Promise.all() batch PATCH to /api/roadmap
- Layout: CSS Grid gridTemplateColumns: repeat(3, minmax(0, 1fr))



2.11 Trade Intelligence Dashboard

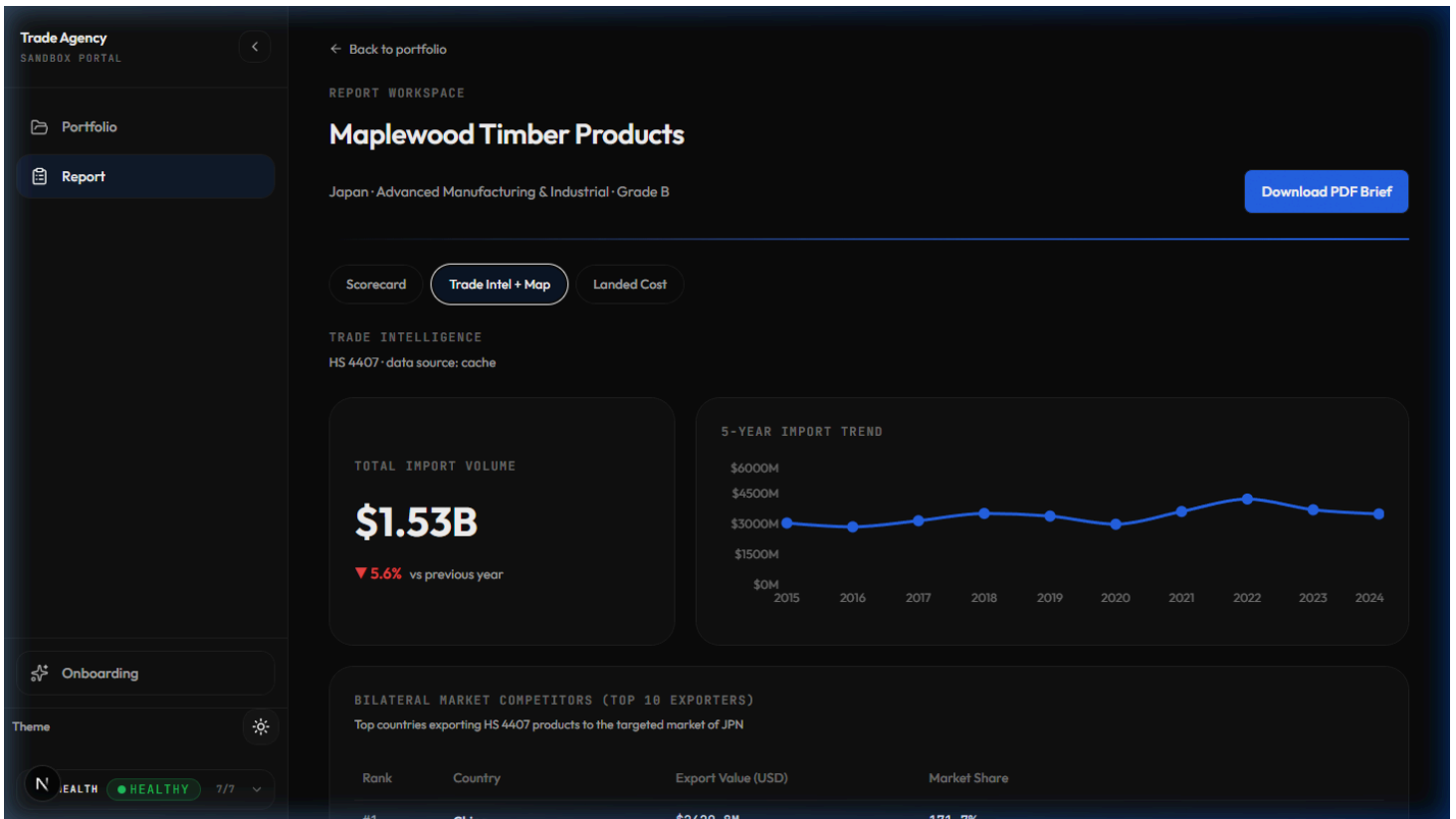
Business Problem: Advisors need UN Comtrade-level trade data to understand market dynamics — import volumes, growth trends, competitive landscape, and seasonality — without leaving the portal.

What it does:

- **Hero metric:** Total Import Volume with YoY change indicator (▲/▼ with percentage)
- **5-Year Import Trend:** Line chart showing import value trajectory (2015–2024)
- **Bilateral Market Competitors:** Table of top 10 countries exporting the same HS code to the target market, with:
 - Rank, country name, export value (USD), market share percentage + bar visualization
- **Import Seasonality:** Bar chart showing monthly volume index
- Data source badge: cache , live , or unknown

Technical Details:

- Component: TradeIntelDashboard.tsx
- Charts: recharts v3 — LineChart , BarChart , ResponsiveContainer , Tooltip
- Data: Three parallel API calls to /api/sandbox/comtrade :
 - type=summary — aggregate volume and YoY
 - type=partners — bilateral partner ranking
 - type=trend — multi-year time series
- Cache layer: lib/comtrade-cache.ts for reducing API calls
- Formatting: formatCurrency() for millions, formatCompact() for billions





2.12 PDF Brief Generator

Business Problem: Advisors need to produce professional PDF reports for their SME clients or for internal review. The report should be downloadable with a single click.

What it does:

- "Download PDF Brief" button in the report workspace header
- Loading state: "Generating PDF..." with disabled button
- Downloads as `export-readiness-<assessmentId>.pdf`
- Error handling with inline error message display

Technical Details:

- Component: `PdfBriefGenerator.tsx`
- Backend: `POST /api/report/pdf` with `puppeteer v25` for server-side PDF rendering
- Flow: `POST` → receive blob → `URL.createObjectURL()` → programmatic anchor click → cleanup
- Error: Parses JSON error body for user-friendly messages

2.13 Landed Cost Tab (Dedicated View)

Business Problem: Some advisory sessions focus entirely on cost modeling. The dedicated Landed Cost tab provides expanded layouts for the solver and margin validator, alongside the full readiness scorecard.

What it does:

- Larger Landed Cost Solver (8-column span) paired with Margin Validator (4-column span)
- Full Readiness Scorecard below (Score Gauge + Pillars Matrix + Critical Gaps + AI Advisory Summary)

Maplewood Timber Products

Japan · Advanced Manufacturing & Industrial · Grade B

Download PDF Brief

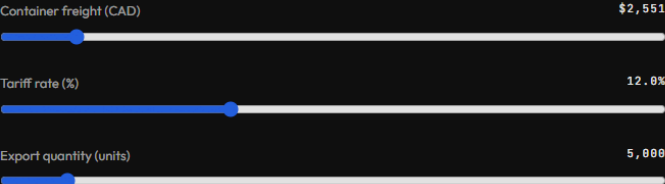
Scorecard

Trade Intel + Map

Landed Cost

LANDED COST SOLVER

Japan · HS 4407 · market inputs: live



Landed cost / unit \$144.56

FX-adjusted margin 23.9%

Margin (23.9%) falls below target (35%) after FX buffer (2.00%).

Cost breakdown

FX buffer 2.00% · volatility fallback

MARGIN VALIDATOR

Profitability gate for Maplewood Timber Products

Margin validation needs review

Unit price covers landed cost
\$49.79 headroom per unit.

Meets target profit margin
Target 35% · FX-adjusted 23.5%

FX volatility buffer applied
2.00% buffer from 30d volatility

READINESS SCORE



Developing

PILLAR MATRIX

Linear readiness breakdown across nine export pillars



CRITICAL GAPS

Pillars below 50% readiness threshold

No critical gaps — all pillars meet the 50% threshold.

AI ADVISORY SUMMARY

Maplewood Timber Products is moderately ready for Japan (Grade B). Core readiness is stable, but cultural adaptation and local standards compliance (JAS certification) require attention.

Onboarding

Theme

HEALTH HEALTHY 7/7

3. Tech Stack Summary

Frontend

Layer	Technology	Version
Framework	Next.js (App Router)	15.5.20
Runtime	React	19.1.0
Language	TypeScript	5.x
Bundler	Turbopack	(built-in)
Styling	Tailwind CSS + custom CSS variables	3.4.17
Typography	Outfit (sans) + JetBrains Mono (mono)	Google Fonts
Theme	Dark-first with light mode toggle	data-theme attribute
Icons	Lucide React	1.24.0

Visualization & Interaction Libraries

Purpose	Library	Usage
Charts	recharts v3.9.2	Line charts (5yr trend), Bar charts (seasonality), tooltips
Geo Map	@visx/geo v4	Mercator world projection for EDC risk map
Map Zoom	@visx/zoom v4	Pan/zoom/reset on the country risk map
Topology	topojson-client v3	Parsing world-atlas TopoJSON → GeoJSON features
Animation	motion (Framer Motion) v12.42.2	Spring transitions, staggered entrances, layout animations
Drag & Drop	@dnd-kit/core v6 + @dnd-kit/sortable v10	Roadmap timeline kanban board

Purpose	Library	Usage
CSS Utils	clsx + tailwind-merge + class-variance-authority	Conditional class merging

Backend / Data Layer

Layer	Technology	Purpose
Database	Supabase (@supabase/supabase-js v2, @supabase/ssr)	Primary data store for SMEs, assessments, notes
Fallback DB	lib/csv-db.ts + lib/in-memory-store.ts	CSV-backed and in-memory stores when Supabase is unavailable
AI	@google/generative-ai v0.24.1 (Gemini)	AI advisory summaries in assessment reports
PDF Gen	puppeteer v25	Server-side headless Chrome PDF rendering
Validation	zod v4	Schema validation for API payloads
Date Handling	date-fns v4	Date formatting and manipulation

External Data Sources

Source	Endpoint	Data
UN Comtrade	/api/sandbox/comtrade	Import volumes, bilateral trade partners, multi-year trends
Tariff Database	/api/sandbox/tariffs	HS code tariff rates by country
Freight Rates	/api/sandbox/freight	Container shipping rates by route
FX Rates	/api/sandbox/rates	Exchange rates + 30d/90d volatility
Sanctions List	/api/sandbox/screen	SEMA / Consolidated Screening List entity check

Source	Endpoint	Data
EDC Risk Tiers	Local data (<code>country-risk-data.ts</code>)	28 country risk classifications

Infrastructure

Concern	Implementation
Health Monitoring	<code>/api/health</code> — checks 7 subsystems, exposed in sidebar badge
Authentication	<code>lib/auth.ts</code> + Supabase Auth middleware
Environment	<code>.env.local</code> with <code>lib/env.ts</code> validation
Testing	Vitest + Testing Library + MSW + Playwright
CI	GitHub Actions (<code>.github/</code>)
Container	Dockerfile for production deployment

4. Page Flow Summary

